

The European Union (EU) membership, since 2004, has been instrumental in modernizing many of the country's policies and administrative structures. Poland also has been one of the largest recipients of the EU's structural funds for roads and highways, urban development, environment, and, to a much lesser extent, water resources management.

Most dikes systems and much of the river infrastructure date back to the beginning of the 20th century.

The catchment areas of the Odra and the Upper Vistula (comprising about one-third of the total Vistula basin) together cover 168,580 km² or 54 percent of the Polish territory, underscoring the strategic significance of the proposed project.

In 1997, the Odra 2006 Program and Law were developed to ensure the protection of the Upper and Lower Odra against 1-in-100-year floods, or better. To date, this program has helped restructure the administrative tools for flood protection and has funded major investments in a variety of measures and infrastructure to achieve the specified protection levels.

The World Bank, the Council of Europe Development Bank (CEB) and the European Commission (EC) have (co-)financed several of these initiatives.

This project experienced a slow start in 2007 for a number of external reasons, including introduction of the new Polish policies and administrative structures after EU accession and the newly imposed regulations. The project is currently performing satisfactorily with to date full commitment of funds for the total cost of about US\$1 billion. The project is also credited with having developed the institutional administrative structure and capacity in south-west Poland to implement very complex hydraulic works for which the cooperation of four ministries¹ and about 24 local governments and agencies¹, and four financiers is required.

In March 2014, the government requested the Bank to provide support for the preparation of a second initiative of national importance with the strategic aim to further complete the protection of all the most vulnerable areas in the Odra basin, and implement a first set of measures to start providing the same level of protection for the Upper Vistula basin. The selection of the activities and investments would follow exhaustive analysis of the priority nature of all possible investments based on integrated basin planning, flood risk analysis, and readiness and no-regret nature of the investment. This new project would also create the platform to start mainstreaming, at national scale, the lessons learnt in policy and institutional development during the Odra River Basin Flood Protection Project.

With its accession to the EU, Poland has had to reform the policies and institutions in its water resources sector to comply with the EU acquis, in particular the Water Framework Directive (WFD) and the Flood Directive (FD).

Tasks on water and flood management are assigned by territory and divided between the Regional Water Management Agencies (RZGWs) and the Boards of Amelioration and Hydraulic Structures of a Voivodship (ZMiUWs). The RZGWs are typically responsible for the main water bodies and rivers within an hydraulically coherent region, such as a large part of a river basin; they operate under the National Water Management Agency (KZGW), which is a semi-independent body under the Ministry of Environment (MoE), and

¹ Namely, the Ministry of Environment (MoE) for implementation, the Ministry of Administration and Digitization (formerly Home Affairs) for the flood protection budget, the Ministry of Finance (MoF), and the Ministry of Infrastructure and Regional Development for coordination of Borrower EU funds.

The Institute of Meteorology and Water Management (IMGW) is responsible for meteorological forecasting and generic modeling of precipitation and run-off.

The new Polish legislation requires the development of integrated river basin management plans (RBMPs) for each river basin. These plans are still under development and have followed the format of the Updated Master Plans, complemented with Flood Hazard Maps and Flood Risk Maps.

The Updated Master Plans were endorsed by the EC as intermediate plans in October 2014 to provide the prioritization of all measures based on a broad feasibility analysis.

The implementation of the complete Vistula flood protection program may well take two to three decades, and will require numerous planning and consultation rounds to select and design the investments and measures

The proposed project targets measures that combine a high cost-effectiveness and a small, or at least, manageable ecological and hydraulic footprint.

The EC and the prospective project co-financiers have confirmed their preference for the Bank leadership of the new large investment program, with the partial aim to use the program to bring the institutional innovations into practice.

Many of these centers have been located along transport corridors and, effectively, in former floodplains and in Poland's most productive agricultural area, with high added-value from orchards and horticulture.

A total of 122,042 people are currently living in the actual flood zones of the 1-in-100 year floods and will receive full or improved physical protection benefit compared to the current situation.

and economically productive areas and assets such as factories, commercial centers, touristic locations, strategic transport and communication networks, farms, etc.

The proposed project will further strengthen the national institutional capacity for flood management and forecasting, as well as improve the capabilities to operate existing and new infrastructure. This is done through more advanced weather forecasting equipment and flood simulation models that will inform decision-makers faster and more reliably regarding the need to evacuate and other precautionary measures to reduce loss of life and economic losses during emergency situations. The project will also help to strengthen the capacity to prepare RBMPs and investment prioritization plans, improve the flood policy framework, and enhance the capability for more reliable forecasting, early warning, and infrastructure operation of existing and new infrastructure. The latter will extend over the whole of south and west Poland and enhance protection levels to an overall population of over 38 million.

These indicators gauge the institutional and administrative capacity through the effectiveness of the system to avoid major impact of flood events and mount operational responses during a

Physical protection benefits include avoidance of, among others: evacuation; inundation of house, factory or other asset to a depth of higher than 0.5m, and avoided loss of life and disease

This effectiveness results from a sequence of proper actions that can happen only if capacities are in place within the RZGWs, ZMiUWs, KZGW, and IMGW: (i) early and reliable forecasting (well-equipped forecasting and run-off and hydraulic routing models); (ii) flood event development that confirms adequate design and construction criteria and decisions; (iii) operation centers that are equipped and staffed, and that are activated timely; and (iv) appropriate decision making to mitigate the flood impact. It is unlikely that a flood crisis will emerge immediately after the completion of the project. Therefore, a separate assessment at the end of the project will be undertaken to determine any improvement in the quality of the procedures and in the ability of staff to manage flood risks.

The proposed project will build on the lessons learned in the ongoing Odra Basin River Flood Protection Project.

The activities will include the reconstruction of dikes and other bank protective works (revetments, parapets, and so on), dredging in the Odra River as well as in canals and the harbor of Szczecin, and river training works, that is, the recalibration and (re)construction of groynes and lateral submerged dams in the river, restoration of bends, and protection of bank. In addition, five bridges need to be raised to facilitate safe passage of icebreakers, and navigation and mooring facilities need to be expanded. A key activity concerns the revitalization of the Miedzyodrze wetland, upstream of Szczecin harbor, to help accommodate water surges and, at the same time, restore some of the ecological and touristic functions of the habitat.

The works comprise of (i) the reconstruction and extension of dikes and embankments along the Vistula to replace old unreliable dikes; (ii) bank stabilization and strengthening with rip-rap, revetments, etc.; (iii) construction of identified dry polders and overflow areas (along the Serafa tributary, and the Raba tributary) to increase upstream water retention; (iv) interventions for river training; and (v) the adjustment of existing weirs and barrages to pass larger flood wave. Through Component 4 additional support will be provided for the preparation of the main parts of the RBMP and the investment prioritization plan for the Upper Vistula.

This component will selectively support the strengthening of institutional capacity in priority areas by: (i) improving the emergency preparedness along the main rivers and their tributaries in south and west Poland by enhancing the forecasting and operational water management capacity; (ii) strengthening the procedures and capacity to prepare RBMPs and investment prioritization plans that are compliant with the EU WFD and FD; (iii) strengthening impact monitoring; and (iv) enhancing the communication capabilities.

and flood protection impacts, and on citizen engagement.

of flood impacts

As part of this, funding will also be provided to facilitate peer-to-peer dialogue on IRWM with appropriate EU member states. A national communication strategy on flood risks and their management will also be developed.

The operations centers are control rooms that will mine forecasting data, simulate likely run-off scenarios, support early warning and decision support processes for emergency response, and operate infrastructure such as weirs, reservoirs, and dry polders to manage the containment and release of flood waves

The implementation of the project included the detail designing and the updating of safeguards arrangements, and took place at an opportune time, as the country was acceding to the EU and had to start implementing the WFD and other directives.

The Odra River Basin Flood Protection Project is co-financed by the CEB and the EC (Cohesion Funds) which provide parallel funding within the framework of the program and investment design parameters set by the Bank.

The cooperation has proved productive, with the co-financiers joining supervision missions and participating in key decisions. With the Cohesion Funds constrained in utilization, the CEB and Bank funds have been instrumental to maintain liquidity and furnish bridging funds, thus leveraging maximum absorption of the EU grants. Without this co-financing arrangement and flexible financial management (FM) by the Bank, the CEB, and the PCU, no Cohesion Funds could have been absorbed.

This compares favorably with other large hydraulic investments such as the construction of the Swinna-Poremba reservoir that after 25 years is not completed yet.

Finally, it is universally accepted that there is a need to move from an exclusive infrastructure approach to a comprehensive and holistic approach to flood management combining infrastructure investments with management and policy measures

Integrated flood risk management should be integrated into comprehensive water resource management plans taking into account the socioeconomic realities, water uses, and vulnerabilities and risks at the river basin level. Thus, successful flood management should combine both structural and non-structural measures.

Also, this lesson is embedded in the project design that prioritizes the creation of upstream storage capacity, the restoration of wetlands, and the improvement of forecasting and management capacity, over the construction of local dikes

The project requires the close coordination of four ministries: The MoE for implementation, the Ministry of Administration and Digitization (MAD, formerly Home Affairs) for the flood protection budget, the Ministry of Finance (MoF), and the Ministry of Infrastructure and Regional Development for coordination of Borrower EU funds. Other key national-level agencies are the National Water Management Authority (KZGW), the Institute of Meteorology and Water Management (IMGW), the National Fund for Environmental Protection and Water Management, and the State Rural Property Agency.

These government officers and agencies decide on environmental, water use, and construction permits, manage the land acquisition processes, and issue important permits for, for example, the use of roads for truck transports. T

The overall strategic coordination, guidance, and quality control will continue to be delivered by the Inter-Ministerial Project Steering Committee (PSC), chaired by the MAD and the MoE, and in which also the MoF and the Ministry of Infrastructure and Regional Development participate. At the operational and day-to-day level, the coordination and quality control will remain the responsibility of the Project Coordination Unit (PCU) that is currently carrying out the same task for the Odra River Basin Flood Protection Project.

This arrangement has proved effective in the ongoing Odra River Basin Flood Protection Project.

Seven new prospective implementing agencies will be added and their capacities will need to be strengthened; this is planned to be done through the existing national systems of coordination and supervision.

Although the preliminary assessment suggests that these agencies are technically and managerially capable, the technical capacities, as well as their capacities with respect to World Bank procurement, FM, and environmental and social safeguards will need to be further strengthened.

The PCU will submit quarterly reports in an agreed format to the MoE, the Project Steering Committee and the Office of Natural Disaster Recovery (ONDR), and the Bank no later than 45 days after the end of each quarter. The quarterly reports would cover, among others, the progress and expected completion date for civil works and goods contracts (both physical and financial progress with photographic evidence), progress and expected completion date for key consultancies, compliance towards environment and social safeguards (including implementation of key features of the environmental management plan (EMP) and resettlement action plans (RAPs)), progress on institutional components, progress toward indicators given in the results framework, training and studies, and activities of the PCU consultants.

An Implementation Completion Report (ICR) will be submitted to the Bank no later than six months after the closing date.

A dedicated survey platform will be developed to allow impact evaluation of floods and flood mitigation efforts, to better analyze the disaggregated impact of the project as well as investment programs (disaggregation by income, location, social variables, gender, and so on) and enable better targeting of future investments and measures. As was done under the ongoing Odra River Basin Flood Protection Project, the PCU will also prepare civil works monitoring tables to provide a more detailed account of progress under these contracts. These tables would provide basic contractual information, progress indicators (both with respect to interim payment certificates and paid invoices), and color-coded task progress.

This extensive analysis and prioritization have allowed selection of the high-priority works and measures out of a long-list of possible identified needs in flood management.

Secondly, the experience with the ongoing Odra River Basin Flood Protection Project has shown that the government is committed to provide the budgets for investment as well as for operation and maintenance (O&M). The field inspections in the territories of the prospective PIUs of the project demonstrate that in general adequate resources are provided for the maintenance and functioning of the infrastructure, recognizing that current O&M budget levels cannot compensate for the funding shortages in the second half of the 20th century. However, with the expansion of the infrastructure, the need to increase the O&M budget is growing apace. As part of the annual work program planning process, the Borrower will include O&M allocations to ensure adequate budgets for the works under the proposed project.

The risk pertains to the distributed nature of flood projects, requiring active commitment and involvement of several national ministries and numerous local governments at voievode, county, and municipality levels

Furthermore, the national scope, the size of the project, and the desire to spread the risks, require the composition of a financing consortium; of these, the EC-related institutions (Directorate General Regional Development) have stated that they will accept Bank guidelines to be applied.

This risk is mitigated by maintaining the conducive and productive relationships that have been developed during the implementation of the ongoing Odra River Basin Flood Protection Project.

The second key substantial risk is the technical quality of the programs for investments and managerial measures for flood risk mitigation that need to be technically adequate and cover the large hydraulically coherent parts of the river basins. In particular, a risk exists that investments are proposed that extend protection to local populations at the expense of downstream population and/or that are not of high-priority or no-regret nature.

This risk is mitigated, partly, by the following steps: (i) the extensive screening of investment proposals that has taken place over the past years by the prospective Implementing Agencies; (ii) the requirement that, for local situations where flood defense could be achieved through different alternatives (for example, through a flood by-pass, higher dikes, local overflow areas, or combinations) the alternatives have been exhaustively compared; and (iii) by the availability, since early 2014 of the draft Flood Risk Maps, that form the basis for the Updated Master Plans that have been completed in 2014 and re-confirm the prioritization of the project selection.

These risks can be mitigated as measures from the on-going project will continue to be used.

Substantial capacity has been built with these entities in the proper application of Bank guidelines in procurement, financial management, and safeguards

As such, the PCU has acted as (i) overseer of the PIU works programming and financing, (ii) the single intermediary between the PIUs, the Bank, the steering committee, and the co-financiers, and (iii) the body to provide clearance to designs, tender documents, environmental and other safeguard documents, and so on.

Based on the findings of the assessment, an economic analysis has been prepared following standard Bank procedures, as well as those commonly applied for investments that would be eligible for the EU Cohesion Funds.

Flood management is preeminently a public good with spatially distributed effects at the local level and at locations elsewhere in the river basin (externalities). In the EU, as well as in all mature market economies, it is one of the larger tasks in infrastructure and water service provision of national and local governments.

The private sector generally contributes to flood protection only through the insurance services to compensate households and businesses that could be affected.⁶⁰ In Poland, the private sector participates intensively in flood projects through the assignments for design, consulting, contracting, and maintenance of the infrastructure.

The economic rate of return (ERR) of the project is 13.5 percent. The Net Present Value (NPV) of the overall project is estimated at PLN 4,815 million (with 5 percent as discount rate) and at PLN 926 million (with 10 percent as discount rate)⁷¹.

The estimated ERR is robust and not very sensitive to costs and benefits: even a 30% increase in investment costs, or up to 40% percent reduction of benefits from avoided material flood damage would maintain an ERR above 10 percent.

The main benefits quantified for this assessment are related to (i) the reduced material and non-material damages from flood hazards; (ii) induced economic activities including mining and agriculture; (iii) enhanced navigability of the Odra River; (iv) increased tourism due to improved safety conditions in the area; and (v) jobs created during the implementation of investments. For estimating the material benefits from flood protection, the area inundated for a sequence of flood events with progressively infrequent return periods were prepared. Hydraulic simulation models were used to provide water levels for floods and duration with different return periods.

The actual damages of the 1997, 1998 and 2010 floods, together with the modeled areas of various land use categories that were inundated, were used to derive unit rates on a per square meter or a per hectare basis for each land use category. The methodology allowed estimating potential flood losses for each area at risk to be protected from flooding and in each of the predicted probabilities of an event.

Due to the complex hydraulic and hydrological processes causing the high water levels, flood management generally necessitates an array of physical and programmatic measures and activities that function in complementarity and are integrated into one management strategy. While a local dike can indeed help mitigate the risk of a local flood, principles of equity and cost-effectiveness impose an integrated approach and strategy at the scale of the complete river basin, in order to avoid protection in one (upstream) location displacing the risk to a different location along the same river (usually, downstream).

In the Middle and Lower Odra, several natural areas alongside the river still give the river space to expand; immediately upstream of Szczecin the vast Miedzyodrze wetland, situated between the two parallel Odra channels, fulfill the same function. On the other hand, the studies on the Vistula, being a larger and more complex hydrological system, will need to be continued to identify all required and cost-effective options. The project would fund investments in medium-scale overflow polders that are located in strategic locations, but will not yet provide the final required space for flood wave absorption.

Over the past decade, under the guidance of the EU WFD and FD, numerous studies have been conducted based on the latest terrain survey technologies and flood risk maps.

The appropriateness of the selection of investments, of the conceptual designs and, where available, the detail designs, and the planning and implementation aspects, were reviewed by three independent senior experts including members of the International Dam Safety Panel for the ongoing project, reflecting expertise in hydraulic engineering, water resource engineering, geotechnology, and construction engineering. They endorsed the project and its components, and raised technical issues that will need to be taken into account as the detail designs and bidding documents are prepared.

The FM arrangements that have been developed under the ongoing Odra River Basin Flood Protection Project will constitute the core for the new project.

The PCU will consolidate the project financial information from all PIUs using an online reporting system and produce full sets of quarterly un-audited interim financial reports (IFRs) that will be submitted to the Bank within 45 days of the end of each calendar semester, from the first disbursement and throughout the project life.

To facilitate the internal administration of financial flows in the new project, the share of the IBRD funding in the overall proposed project cost would be determined on an annual basis, through an Annual Work Plan (AWP) prepared by the Borrower, and agreed between the Borrower, the Bank and other co-financier partners involved in the proposed project implementation. Therefore, the percentage of financing from the IBRD loan may be different each year, depending on the availability of funds from other funding sources in each implementation year.

The MoF will open a separate Designated Account (DA) in the National Bank of Poland. The World Bank loan proceeds will flow to the DA and from there will be transferred to the PIUs accordance with the existing budgetary procedures applicable for foreign sources of funding.

The capacity of the new implementing agencies that have not been involved in implementation of the ongoing Odra River Basin Flood Protection Project will be strengthened based on close supervision, coordination, and training provided from the PCU and existing PIUs, as well as additional joint fiduciary training provided by the Bank.

The selection criteria and technical design of all project activities considers the minimization of adverse impacts, including the need for involuntary resettlement. In many cases the works involve the rehabilitation of existing infrastructure in mostly government-owned lands located in sparsely populated peri-urban and rural areas (e.g. rehabilitation of existing dikes and groynes), for which land impacts are none or limited.

It must be noted that no project activities requiring significant resettlement will be supported by the project.

Gender: Given that the Project will provide flood protection to the general population in the Odra and Vistula river basins no gender-differentiated benefits and/or impacts are expected.

At the broader level, the preparation of the River Basin Management Plans (RBMPs) and prioritization of investments will be based on multi-stakeholder consultations (e.g.

At the sub-project level, the location and design of the investments will also be subject to public information and consultation through the corresponding municipalities.

It is important to note that the majority of the investments concern rehabilitation and modernization of already existing structures.

Thus, from a project implementation perspective, a high level of environmental protection is being applied (e.g.

Besides strictly applying the selection criteria for those investments, the site-specific Environmental Management Plans (EMPs) will emphasize the reduction and mitigation of potential negative impacts, chiefly during the construction phase. The activities within protected nature areas will be limited to the restoration of the existing linear flood defense infrastructure and, in a small number of cases, to local dredging aiming to restore the natural flow of water.

The project will also finance the construction of several overflow areas (dry polders). Since the specific locations of these have yet to be finalized, the project will ensure the application of the investment selection criteria during the feasibility study and conceptual design stages and require the polders to be carefully evaluated with respect to location, size and impacts, to ensure that possible adverse effects are minimized and adequately mitigated. Overflow areas and/or polders with major negative impacts will be analyzed further before suggesting appropriate environmental mitigation measures and/or change of location. Any of the structures that may have a major negative impact on environment, or structures that can be categorized as environmental category A, will not be eligible for financing under the Project.

Beside the project-scale ESMF, separate site-specific Environmental Management Plans (EMPs) will be prepared for all investments once identified. The preparation of preliminary information for some EMPs is underway for the priority pre-identified project activities that could be ready for tendering in 2016 following their confirmation. These EMPs will become essential components to the bidding documents and subsequent construction contracts.

OP 4.04 on Natural Habitats: Based on available information, the project will have significant positive environmental impacts in terms of protecting floodplains and aquatic ecosystems.

Appropriate provisions for archeology investigation and rescue works near these buildings, and for chance finds have been included in the ESMF.

OP 4.37 on Safety of Dams: Although the final selection of polders to be rehabilitated or constructed has not been made, it is known that some of the polders will include construction of dams and weirs.

Therefore, the project retained the independent Panel of Dam Experts that already exists for the on-going Odra River Basin Protection Project and is adjusting its terms of reference

The design of all dry polders under the project will be prepared in compliance with the Safety of Dams policy.

In parallel, the trilateral Odra River Commission addresses the technical aspects of the river issues of common interest. The Odra Commission has agreed on the technical specifications of flood protection along the Border Odra.

Project Development

PDO Statement

The project development
Vistula River basins and

These results are at

Project Development

Indicator Name	Core
Total area in the 1% flood plain benefiting from enhanced protection and operational forecasts	

⁸ Based on current 1% flood

⁹ Estimated for horizon to `

Total population benefiting from enhanced protection and operational forecasts (gender disaggregated)	
Flood Operation Centers established and functional	

Intermediate Result

Indicator Name	C
<i>Component 1A:</i> Length of enhanced protection in the	

¹⁰ Based on current flood n

¹¹ Estimated for horizon to

Zachodniopomorskie Province
<i>Component 1B:</i> Modernizing and reconstructing the middle and lower Odra river systems to upgrade to a Class III waterway
<i>Component 1C:</i> Extension and construction of flood embankments to protect Słubice City
<i>Component 2A:</i> Construction of dry polders in the Nysa Kłodzka Valley
<i>Component 2B:</i> Length of enhanced protection in the Nysa Kłodzka Valley

¹² This component represents

<i>Component 3A:</i> Construction of dry polders to protect Upper Vistula towns and Kraków	
<i>Component 3B:</i> Length of enhanced protection for Sandomierz and Tarnobrzeg	
<i>Component 3C:</i> Length of enhanced protection for the Raba sub-basin	
<i>Component 3D:</i> Length of enhanced protection for the San sub-basin	
<i>Component 4B:</i> Flash flood systems for sub- basins operational	

RBMPs or Investment Prioritization Plans drafted for Vistula sub-basins	
% of project- supported RBMPs and investments informed by citizen feedback through consultations (disaggregated by gender)	

In Western Poland, the Odra benefited from ample government attention starting in the 19th century and this had led to major investments in river navigability and flood protection in the early 20th century.

After 1945, investments in flood protection took a back seat as the country was rebuilding its cities and its industrial base. Currently, flood protection is a high priority in government policy and has broad political support across all parties.

In response, the government established, in 1997, the ONDR in the Office of the Prime Minister; to date the ONDR manages the sizeable budgets directed at flood management and also the budgets that have co-financed the ongoing Odra River Basin Flood Protection Project. In 2005, the ONDR was placed in the (then) Ministry of Home Affairs (now, the Ministry of Administration and Digitization).

In addition, the Parliament also adopted a law in the summer of 2010 to streamline the land appropriation procedures for investments in flood protection, in line with the procedures for highway construction. The government policies call for a long-term program with institutional strengthening and extensive investments to address the flood risks.

In parallel, the Odra River Commission, comprising senior representatives of the Polish, German, and Czech authorities, addresses all technical aspects of the river issues of common interest. It sets the standards and aims for flood protection, river management, and commercial shipping.

The main flood protection measure consists of passive dike systems for which the crest height can be calculated so that it can withstand all high-water waves smaller than the one that is calculated to recur once in, for example, 100 years (1 percent chance of occurrence per year), which is a common minimally desired protection level for most urbanized areas and compliant with EU directives.

The most cost-effective technology to demolish ice dams in a controlled way is by icebreaking ships that cut the ice preventively before it forms thick aggregates.

The Polish government has the objective to develop its national expertise and institutions to better cope with the three flood types.

After intensive studies carried out by German research institutes, the Odra Commission had agreed in the early 2000s on the technical specifications of flood protection along the Border Odra. Investments and measures under the project in and along the Border Odra will assist Poland to meet these agreements

The data are drawn from various studies and databases, mostly from the hydrological data collected and analyzed by the IMGW.

Because of the fractured nature of the Upper Vistula, a comprehensive and spatially integrated approach to flood protection as part of flood risk management was only proposed during a study designed to prepare the framework and tasks of The Upper Vistula River Basin Flood Protection Program which became a government document (Resolution No. [REDACTED])

[REDACTED] It is oriented toward structural and nonstructural flood protection measures which are designed to gradually increase the level of flood safety of people as well as economic and cultural assets in the Upper Vistula basin with a time horizon of 2030.

Also, analysis of the flood waves (Hydrographic Analysis of the 1997 Flood on the Upper Vistula, IMGW, Warsaw, 1999) confirms the hydraulic disconnect between the Upper, and the Middle and Lower Vistula segments.

[REDACTED] It also received access to large EU funding allocations (Structural and Cohesion Funds) that can finance up to 85 percent of eligible infrastructure and other measures

[REDACTED] To transpose the regulatory acquis of the EU, an umbrella-type Water Law was enacted, and the preparation of River Basin Management Plans (RBMPs) started.

[REDACTED] The Voivodship ZMiUWs are responsible for all minor water courses and all dike systems that stand apart from the rivers; this pertains to the majority of dikes and embankments.

Firstly, a new Water Law is currently proposed for adoption in Parliament to streamline the water management administrative structure and update a number of regulatory features, for example, on the preparation of the RBMPs.

As it was recognized that possible approval of the RBMPs by late 2015 or 2016 could jeopardize the newly proposed Odra-Vistula Flood Management Project, the EC reviewed and endorsed the interim Updated Master Plans for the river basins in October 2014. The EC agreed to the principle that many proposed investments and measures are of no-regret nature, and can be implemented with due safeguards and proper cautionary preparation, while other investments and measures have long-range and complex impacts that affect the design, operation, and integrity of other defense structures as well as other land use and environmental features in the basin, and, therefore, these need to be assessed within the context of a full-fledged basin-wide analysis and plan.

The Annex List 2 comprises 450 items that are considered complex and/or subject of decisions that require a basin-wide analysis; these investments and measures can potentially be eligible for EU funding after approval of the RBMPs that will need to be resubmitted after 2015.

Flood Management Investments and Measures Prioritization

16.

was agreed upon.

For the project, a selection of priority investments and measures

The maps are based on new-generation digital elevation maps created from remote sensing and LiDAR surveys; they are prepared for return periods of 10, 100, and 500 years.

Thus, this flood hazard information needs to be complemented with both ground-truthing to ensure that the simulation outputs reflect actual observed flood extents and with an analysis of damages and losses. The project will also be

¹ Flood damage relates to the physical and non-physical costs for reconstruction and repairs of infrastructure and other assets to restore livelihoods to the situation before the flood. Flood losses comprise the costs to the economy resulting from disrupted economic activities, notably loss of income of businesses and households.

upstream retention, generally, can also lead to lower investments downstream

Investments in

This component will also allow to maximally benefit from the existing institutional structure for project implementation created by the Odra River Basin Flood Protection Project.

It is expected that the results of this component will assist Poland to meet the international agreements on the Odra management

In addition, in the sub-basins of the tributaries

¹ No regret measures are those that are characterized by the following: (i) having only local impact, hydraulically and environmentally, or, if there is downstream or upstream impact, this is well understood and managed; nor are they themselves impacted significantly by the design of future upstream or downstream investments; (ii) in demonstratively vulnerable areas associated with high benefits from protection measures that are unlikely to be excessively expensive; (iii) without complex interactions across a large part of the basin that would necessitate a robustly optimized basin plan to yield the most cost-effective variant, and (iv) have a comparatively modest ecological footprint and are otherwise non-controversial.

For several proposed works and measures, the economic rationale is confirmed and advanced documentation for works exists, yet for other sets of investments additional studies to optimize the investments will be required before further decisions can be made on their merit

It is to be noted that the Flood Hazard Maps identify vulnerable areas that, however, have not been prioritized for the project.

The Warta was not prioritized because it has a low readiness to proceed, which is the consequence of the presence of the large Natura 2000 protected forest tracts that, for the moment preclude investments in flood management. Developing integrated plans and consensus on the management of the water resources and environmental values in that area will require more years of consultations and studies.

However, because the flood plain is relatively wide in that section and the vulnerable asset values in the surrounding areas are limited because of relatively good natural protection of towns, the area was not considered a priority.

The flood risks along the Odra main stem and its tributaries are well documented and the remedial investments and measures extensively studied. The draft Updated Master Plan and the draft RBMP for the Odra largely confirm the existing long-list of required investments and measures that had been identified before; however, the plan also introduced corrections. On the other hand, the preparation of such a long-list of meritorious investments and measures had to be initiated for the Upper Vistula.

Similarly, the FD requires the preparation of Flood Risk Maps and Flood Hazard Maps. The ensuing flood management measures need to be informed by and integrated in, the RBMPs.

In these instances, the project will support financing for in-depth studies but not the investments in works.

Crossing a mostly rural and forested part of western Poland, the Free-flowing Odra does not need major dike systems, but primarily extensive river training.

The winter floods need to be tackled by the operation of icebreakers that have to operate preventively to avoid the build-up of ice dams.

Firstly, many of the old dike and embankment systems near or around towns and infrastructure need to be modernized and/or expanded.

A one meter high water layer on the wetland would store about 1 billion m³ of floodwater. This rehabilitation will have the additional benefit of restoring the previous ecological conditions with significant improvements of water quality, and of the fish, macrophyte, and bird diversity.

Thirdly, a number of diverse investments along the river are necessary to allow the operation of the new generation of energy-efficient icebreakers

After half a century of neglect, the Middle and Lower Odra River beds need to be thoroughly trained and dredged. To ensure adequate depth along the centerline of the river over long distances of the river, existing and new groynes (short low breakwaters protruding perpendicularly from both river banks) and lateral breakwaters need to be (re-)constructed. The groynes will be redesigned to maintain erosive conditions in the river centerline to reallocate bed loads through some stretches that need to be dredged. The hydraulics of the groyne designs have been studied earlier with both physical and numerical simulation models at various research institutes. The design dimensions to achieve protection levels against 100-year return floods (summer floods) and allow icebreaker operation in the winter, while at the same time avoid creating additional friction under high-water flow discharge condition, have been formally set in standards of the tri-lateral Odra River Commission. The groynes on the German bank have been reconstructed over the past 10 years to the agreed standard. The project, thus, will assist Poland to meet the international agreements on river management and flood protection. As an additional benefit, the river training and bridge elevation will facilitate commercial shipping of larger bulk loads (construction materials and grains) and containers.

The protection investments of communities and other assets along the river is based on a prioritization analysis carried out for the Flood Risk Management Plans that have been carried out in 2014, as well as for the Updated Master Plan for the Odra that was approved in October 2014.

The economic analysis shows that the individual investments (where required, grouped per objective, such as river training) as well as per component, meet the prioritization requirements.

- The works will comprise construction of 2.3 km of flood embankments with overpasses, anti-filtration membrane, provision of mobile flood screen for one location, gabion protection, and ancillary works to protect two towns from Odra water backflow. This task also includes modernization of the Marwicki polder. The works will include the reconstruction of the 3.49 km long dike with protective screens and a top road, as well as the modernization of the Krajnik pumping station.
- .A.2 Flood protection of Ognica village on Odra River. The works will comprise a 300 m long flood embankment with anti-filtration screens and access roads, as well as refurbishment of the Rynica-Ognica pumping station to drain water from the land side of the embankment.

This wetland of 5,200 ha forms the former overflow plain situated between the two parallel arms of the Lower Odra between Gryfino and Szczecin. The works will include the dredging and stabilization of 21 channels; reconstruction of 32 automatic gates, locks, and passes; levelling of 60 km of dike tops; and provision of 32 footbridges. The works will include the development of a hydraulic-ecological simulation model to optimize flow patterns conducive to flood management and ecological productivity.

The task includes the reconstruction of regulating structures and groynes, lateral breakwaters, revetments along banks, bank stabilization, minor dredging, and so on.

.B.2 Modernization works on the Border Odra and Lower Odra (under RZGW Szczecin) to provide good conditions for icebreaking in winter.

.B.3 Construction of docking/mooring infrastructure on the Lower Odra and Border Odra including the new marking of the shipping lane.

This task includes the deepening of the shipping route on the lake and involves dredging of about 1 million m³ of sediment, adjustments to banks and revetments, and so on.

This task also includes dredging of the Klucz-Ustowo ditch. This is a minor shipping canal in the Szczecin waterway that needs to be dredged to allow ship passage.

.B.5 Raising of five bridges to ensure a minimum clearance.

The adaptation of the bridges is agreed under the International Odra Commission; however, for the railroad bridge technical coordination will need to be ensured with German authorities. This has been initiated through the Chair of the Polish-German Border Commission, who is concurrently the plenipotentiary for the Lower Odra flood protection program.

This

subcomponent includes the following proposed works:

1.C.1 Extension and construction of flood embankments.

The top of the embankment will be finished
to allow emergency installation of mobile flood screens.

The flood risk mitigation comprises a combination of upstream dry floodwater retention reservoirs (dry polders), dike systems, and measures to increase the flood wave passing capacity such as widening the channel girth by removing added obstacles and reducing friction, raising of minor bridges, and so on.

Both the PIUs have cooperated in the ongoing Odra River Basin Flood Protection Project and have, thus, a high level of institutional preparedness.

The salient parameters of the reservoir are: maximum capacity of 9.9 million m³; dam length along the crown axis of 735 m; maximum dam height of 19.3 m; reduction of the design flood wave (200-year return period) is 51 percent and the reduction of the control flow (500-year return period) to establish the structural integrity of the dam body, of 20 percent.

The salient parameters of the reservoir are: maximum capacity of 1.9 million m³; dam length along the crown axis of 450 m; maximum dam height of 15.7 m; reduction of the design flood wave (200-year return period) is 76 percent, and the reduction of the control flow (500-year return period) to establish the structural integrity of the dam body, of 54 percent.

As part of the entire work, in particular, the following will be performed: section-based modification and renovation of the existing bank protection measures; construction of new embankments and floodwalls on the section whose total length is 14.5 km; modification of the existing embankments and floodwalls on the section for a total length is 6.5 km; enhancement of throughput of 38 bridge and footpath structures; and enhancement of throughput of 13 small weirs and barrages.

In total, work will be performed on a 26.8 km long river section.

As part of the entire work, in particular, the following will be performed: section-based modification and renovation of the existing bank protective measures and enhancing the throughput of the two river beds; construction of new embankments and floodwalls on the section with a total length of 25 km; modification of the existing embankments and floodwalls on the section whose total length is 4 km; enhancement of throughput of 23 bridge and footpath structures; and enhancement of throughput of 9 minor weirs and barrages.

As part of the entire work, in particular, the following will be performed: section-based modification and renovation of the existing bank protective measures and enhancing the throughput of the beds of the two rivers; construction of new embankments and floodwalls on the section with a total length of 8 km; modification of the existing embankments and floodwalls on the section with a total length of 6.5 km; enhancement of throughput of 66 bridge and footpath structures; and enhancement of throughput of 12 minor weirs and barrages.

Therefore, the potential flood risk (the population number as well as their property and economic asset value, times the hazard) is also higher than the average risk for the whole country. Thus, the prioritization of the Upper Vistula for the project has a sound technical, social, and economic rationale.

The concerted action to prepare a comprehensive and long-term strategy was initiated in 2011 but was catalyzed by, on one hand, the requirement to develop EU-compliant Flood Hazard Maps and Flood Risk Maps, and, on the other, the opportunity to draw lessons from the Bank-financed Odra River Basin Flood Protection Project and extend its implementation structure to the Upper Vistula through the new project.

- The Raba tributary, and, inside that sub-basin, selected coherent investments that were prioritized based on a technical, social, and economic ranking of strings of investments
4. The San tributary, and, inside that sub-basin, selected coherent investments that were prioritized based on a technical, social, and economic ranking of strings of investments

46. Though the Raba and San rank high in priority among all right-bank sub-basins and command priority attention, they may not necessarily yield the highest economic returns when all benefits are calculated across the whole basin and for all possible precipitation and investment scenarios.

The spatial distribution of the flood damage to public assets in 2010 is shown in table 2.5, however, the figures exclude damage to movable assets of private entities and individuals, a part of the damage to municipal infrastructure and the (indirect) losses, that is, those caused by disruption of economic activity, which were exceptionally high due to the two-month persistence period of the flood.

hydraulic works.

Generally, these are competent agencies familiar with large

Thus, it has been concluded that the original embankment height should be essentially maintained, but that the dike bodies need to be rebuilt according to modern standards, and notably should have impermeable screens to prevent seepage and upwelling phenomena.

Therefore, measures need to be taken to readily convey rainwater from the city area to the river system.

This infrastructure stabilizes the river water table within the city under normal conditions but it poses an additional risk under flood conditions, as two of the three barrages have proven unable to cope with the hydraulic discharge under high-water condition, which results in the reduced flood carrying capacity as well as in the loss of stability of the riverbed downstream of the barrages and adverse morphological changes in the riverbed upstream of the barrages.

This task comprises the completion of the passive protection along the Vistula over a distance of 20.95 km, in three sections.

These designs are based on the Concept Study of 2000, and the legal requirements for embankment design.

.A.2 Flood protection in Serafa Valley.

The task will comprise construction of four small dry polders to complement a fifth that has just been completed by the ZMiUW.

The pre-feasibility studies were completed in 2010, based on hydraulic modelling of the flood routes.

The 2010 flood crest reached the top of the embankments but the dike breaches were caused by seepage and collapse of the embankments caused by saturation after an unusually protracted high water condition and the occurrence of a double wave crest (associated with flood waves of 5,264 and 4,722 m³/s, respectively) of which the second proved most destructive

The function and feasibility of large multipurpose reservoirs will need to be the subject of more expansive analysis, and this will, in turn, define the role of new overflow areas and dry polders upstream of Sandomierz.

Therefore, the embankments at Sandomierz will be designed at levels that are comparable to the 1997 and 2010 floods, that is, for a design flood of 5,800 m³/s as these are about the tallest that can be feasibly constructed. The incremental protection to reach the Q1 flood protection of 6,600 m³/s will need to be obtained from upstream flood wave retention. Tentatively, the required storage capacity has been calculated as about 160 million m³, which is also the volume that was spilled in the 2010 flood episode.

.B.1 Flood protection Sandomierz.

The pump station will have a capacity of 1.5 m³/s. Struga A river flood protection including the Nadbrzezie pump station modernization. The Struga A flood protection will extend over a distance of 1.6 km; the pump station capacity will be upgraded to 2 m³/s. Expansion of embankments protecting the glassworks and residential areas in Sandomierz. The length of the embankments will be approximately 2.78 km, with a crest height of 1.67 m.

4. The total embankment length will be 27.3 km, and the dikes will be raised by approx.

Szewce pump station reconstruction This will encompass the construction of a new retention tank extension and new outlet.
Extension of the Zajezerze pump station.

.B.2 Flood protection Tarnobrzeg.

This subcomponent will include a number of works in the Raba sub-basin that are selected using an optimized investment variant (investment scenario) which prioritizes those investments that increase retention of water upstream and/or combine social and technical feasibility with the highest economic returns, that is, those that generate demonstrated benefits at local scale, in the downstream stretch, as well as further downstream along the Upper Vistula.

However, the new calculations show that the reservoir could play a larger role, at low cost, by storing an amount of 20 M m³ of floodwater. This would have, possibly, implications for the safeguarding of the water quality. Similarly, other important active flood retention structures have been identified but the hydraulic impacts of these, and the technical feasibility need to be confirmed.

The Phase 1 investments are those that have been confirmed on technical and economic merit, and whose functioning does not depend on the selection of the other structural elements and other measures (notably, the change in the operational schedule of the Dobczyce Reservoir) in the basin. The Phase 2 investments are those that have been identified as key elements in the most cost-effective and sustainable variant, but still need to be confirmed, as well as their geological surveys and detailed designs completed; this would pertain to 4 dry polders, and 1.6 km of new dike.

The subcomponent will finance the Phase 1 investments, and the full preparation of the Phase 2 investments, to bidding level.

This subcomponent will include a number of works in the San sub-basin that are selected using an optimized investment variant (investment scenario) which prioritizes those investments that increase retention of water upstream and/or combine social and technical feasibility with the highest economic returns, that is, those that generate demonstrated benefits at local scale, in the downstream stretch, as well as further downstream along the Upper Vistula.

Like in the Raba basin, key structural elements for are flood reservoirs and dry polders that have been identified, but the hydraulic impacts of these, and the technical feasibility need to be confirmed.

The Phase 1 investments have been confirmed on technical and economic merit, and their functioning does not depend on the selection of the other structural elements and other measures (notably, the construction of flood retention reservoirs and dry polders) in the basin.

The Phase 1 investments concern the Sanozcek dike (1.61 km), and the construction of the Golcowa dry polder (Ropa River). The subcomponent will finance the Phase 1 investments, and the full preparation of the Phase 2 investments in the three sub-basins, to bidding level.

The Bank has maintained a dialogue with the government and the EC on the institutional reforms and the next steps on the road toward integrated water resources management (IWRM) in Poland. A consensus has emerged that the EC is better placed to carry out a dialogue on the institutional and regulatory aspects of flood management, because of its *acquis communautaire* (the accumulated legislation, legal acts, and court decisions which constitute the body of European Union law) and the requirement that member states transpose the WFD.

The EC intensively monitors and assesses this development, is able to fund capacity building, and disposes of financial incentives to the reform. For example, the EC is the main financier of the national Flood Hazard and Flood Risk Maps, and of numerous studies and TA relating to the legal drafting and RBMP preparation.

Poland, EC and co-financiers have confirmed their preference of technical and operational leadership by the Bank for new large investment programs with the partial aim to use the programs to bring the institutional innovations into practice and pilot initiatives that are less well covered under EC support, such as the conversion of RBMPs into realistic investment programs. The project will therefore focus on (i) the support for the preparation of the RBMP and investment prioritization plans for key sub-basins of the Upper Vistula basin which has a complex hydrology; (ii) enhanced capacity for forecasting and early warning, and for real-time operational management; (iii) strengthened monitoring of impacts; (iv) and strategic communication of flood risks.

The component builds logically on the support to the flood forecasting capacities of Poland extended under the two earlier Bank Odra Projects as well as under other financing from the EC. The ongoing Odra River Basin Flood Protection Project has notably assisted the IMGW to consolidate, on a generic basis at national level, its technical capacity for more reliable and more longer-time forecasts by improving its data base, its data management protocols, and its forecasting routines and Hydrometeorological Monitoring, Forecasting, and Warning System (SMOK). The new project will further strengthen the flood monitoring and warning system for the Upper Odra and for the Upper Vistula at a more operational level.

The component will notably be arranged to ensure that the activities on the forecasting side (IMGW-PIB) and the operational side (RZGWs) will become better integrated and made complementary, avoiding overlaps and gaps. Strengthening of the flood forecasting system and the operations centers will contribute substantially to the flood reducing benefits obtained with the construction of dry polders for flood management in Poland under the current and all previous investment schemes.

The activities will also specifically investigate the type of appropriate software and hardware to improve the reliability and lead time for flash flood warnings in sub-basins that are prone to such events, and to analyze appropriate flood mitigation investments

These subcomponents complement each other and will lead to a fully integrated system which provides IMGW/ IMGW-PIB, in their forecasts, with the benefits of knowledge on the immediate hydraulic structure operations carried out by RZGW, while RZGW will have at any time the most up-to-date information on forecasted river discharges.

achieve this, many of the monitoring instruments require replacement.

The current World Meteorological Organization standards require a higher density of monitoring equipment and upgrading.

To

The improvement of the flood forecasting system under the project foresees the following activities under the subcomponer.^d :

- i. Modernization and expansion of the network of telemetry weather monitoring stations for the needs of hydrological forecasting and modelling. Principal investments include: (a) improvements/replacements at automatic weather stations; (b) replacing tipping bucket rain gauges by more reliable laser gauges; (c) installing snow depth sensors; (d) upgrading radio communication systems; (e) equipping some stations with water quality monitoring facilities; and (f) various installation and construction works.
Modernization and expansion of the system of measurements and forecasts of hydro-meteorological and oceanographic parameters in the waters of the Baltic Sea to enhance the flood protection of the coastal areas. Principal investments include: (a) measurement equipment with sensors and data transmission equipment; (b) maintenance equipment; and (c) hydrodynamic modelling of the Baltic coastal zones.
Modernization of the network of the POLRAD weather radars to increase the effectiveness of the hydrological models). Principal investments include the installation of various fixed and mobile radars, radar tower adaptations, or the new construction of these.
Update and modernization of the Hydrology System (a Management Information System [MIS]), including the development of training, application, and infrastructure.
- v. Modernization and development of an integrated hydrological and hydraulic modelling system as part of the hydrological monitoring, forecasting and warning system. Principal investments include: (a) rainfall-runoff model development; (b) hydrodynamic model development; and (c) training.
Principal investments include: (a) IT platform extension; (b) development of the national flash flood forecasting system; and (c) catchment indexing and rainfall-runoff model development.

Principal achievements of this investment will be the high potential to reduce flood damages in case of emergencies and the prevention of flood damages due to better guidance for icebreaking operations.^m Principal activities of this subcomponent are:

i.

This work package comprises the following subtasks: (a) installation of monitoring and telemetry equipment at reservoirs, dry polders, and critical dam/dike sections; (b) modernization and implementation of local control systems for selected hydraulic structures; (c) improvement of equipment for guiding the inland navigation on the Upper, Free-flowing and Lower Odra; (d) modernization of field/mobile units; (e) installation of equipment at the operations and crisis centers to secure optimal provision of services; and (f) measures to assure reliable functioning and resilience of all essential installations).

Design and development of the informatics platform at the operations centers, including the user interface and the database for the storage, quality control, and retrieval of information supporting the tasks).

Integration of the dry polder and reservoir operation system of RZGWs into the operational forecasting system of IMGW;

v. Training of staff of the operations centers and the IT Department.

The PCU will be restructured to combine the coordination of the ongoing Odra River Basin Flood Protection Project and the new project.

² A secondary benefit is that such system will allow simultaneously to streamline the operational supervision of shipping movements and the rapid response in case of threatening situations such as oilspills.

The support to the implementing agencies will consist of dedicated TA teams that will have the task to, among other things, but not limited to, review and update designs and permit applications; carry out detail designs; prepare procurement documents and supervise contractor procurement; carry out land acquisition and environmental studies; prepare FM documentation; and prepare all reporting.

As part of this, funding will be provided to facilitate peer-to-peer dialogue on IWRM with another appropriate EU member state that is considered to have successfully transposed the EU acquis. A national communication strategy on floods will be developed.

The PCU will be able to recruit experts and consultants, for legal assistance, for additional studies (including to prepare a follow-up project), for the International Dam Safety Panel, and for audits.

The polders are the Gromiec, Jankowice, Smolice, Spytkowice, Kamien, Rusocice, and Klokocyn and represent a total storage capacity of up to 110 million m³.

The positive effect of the investment concerns the lowering of the flood wave crest height by 75, 63, and 40 cm at Denbicki bridge, Grunwaldzki bridge, and Nowa Huta, respectively.

Secondly, the project will support the government with a general strategic study to assess the mid-term and long-term impacts of climate variability and land-use changes on the precipitation and temperature patterns and on the general characteristics of the run-off, hydrology and floodwave propagation. The study would formulate recommendations that would assist the government and other stakeholders to achieve better planning by incorporating such information in the river basin management plans and flood management plans.

The proposed project requires the close coordination of four ministries: The Ministry of Environment (MoE) for implementation, the Ministry of Administration and Digitization (formerly Home Affairs) for the flood protection budget, the Ministry of Finance (MoF), and the Ministry of Infrastructure and Regional Development for coordination of Borrower EU funds.

The overall strategic coordination, guidance, and quality control will be delivered by the Inter-Ministerial Project Steering Committee (PSC)², chaired by the Ministry of Administration and Digitization (formerly, Home Affairs) and the MoE, and in which also the MoF and the Ministry of Infrastructure and Regional Development participate. At the operational and day-to-day level, the coordination and quality control will remain the responsibility of the Project Coordination Unit (PCU) that is currently carrying out the same task for the ongoing Odra River Basin Flood Protection Project.

For all the activities in Components 2 and 4, and for some of those in Component 1, the PIUs will be the same as under the ongoing Odra River Basin Flood Protection Project, providing continuity, and ensuring that lessons from the current Odra River Basin Flood Protection Project will be used in the new Odra-Vistula Flood Management Project.

² PSC will be the same as the one for ongoing Odra River Basin Flood Protection Project.

In addition, the IMGW has implemented a large component aimed at strengthening the national early-warning and data management system to satisfaction. The new Odra-Vistula Flood Management Project will build on the experiences and staff of these two institutions. Seven new prospective implementing agencies would be new to WB-financed operations and their capacities will need to be strengthened; this is planned to be done through the existing national systems of coordination and supervision

The preliminary assessment suggests that these agencies are technically and managerially capable; have a tradition of managing medium to large investments; and have expressed readiness to participate in the proposed Odra-Vistula Flood Management Project.

The technical capacities, as well as their capacities with respect to World Bank procurement, FM, and environmental and social safeguards will need to be further strengthened; this will be monitored closely in particular in the first years of project implementation and additional training and supervision extended, if necessary.

Financial Management, Disbursements and Procurement

Financial Management

F

The FM arrangement of the ongoing Odra River Basin Flood Protection Project will constitute the core structure for the new Odra-Vistula Flood Management Project.

The FM assessment was completed including specification of the next steps related to the upcoming new Odra-Vistula Flood Management Project, including (i) updating of the FM section of the Project Operational Manual and its distribution to all PIUs and (ii) selection of the auditor type: NIK or private auditor. The overall initial FM risk for the proposed project before and after mitigation measures is assessed as Moderate.

The budget for the proposed project, including loan funds and counterpart funding, is included in the state budget and prepared and approved by the Parliament on an annual basis. The proposed project budget is prepared in accordance with the Public Finance Law and classified as a reserve for disaster recovery, including earmarked funding from the loans (the Bank and other IFIs) with co-financing by the national budget.

The loan funds will flow to the DA maintained by the MoF in the National Bank of Poland and from there will be transferred to the implementing agencies employing the existing budgetary procedures applicable for foreign funding sources. The loan funds are transferred from the MoF to the state budget account.

This process of application for funds, and, thereafter, the transfer of funds, is complex as each intermediary step requires a paper decision and a separate approval from the KZGW, the MoE, the Ministry of Administration and Digitization (MAD), and the MoF.

Overall, the internal control systems at the PCU and the PIUs have been enhanced and have proven adequate in the ongoing Odra River Basin Flood Protection Project, and it relies on the existing application of the Public Finance Law requirements including the requirement for efficient and economic use of the funds. The contracts to be financed from the proposed project sources will be included in the Procurement Plan to be approved by the Bank.

The funds for this proposed project are included in the state budget in the so-called special reserve which is used for all external loans. Implementing agencies will apply for the proposed project funds including loan and counterpart funding from the ministry in line with the budgetary procedures but with an additional acceptance by the PCU.

The PCU will continue to monitor and coordinate the flow of funds including the monitoring of the complex flow of funds from the MoF to the implementing agencies, managing the cash flow liquidity of the proposed project by consolidation of the cash forecast, consolidation of the information about the use of the funds, and preparation of the reports and records for documentation of the expenditures to the Bank.

Usually, contract monitoring is done in Excel and includes contract deliverables, unit/component price (if applicable), total amount of the contract, invoiced and paid amounts, and commitments.

The contract monitoring is performed by the technical manager and/or by the financial staff in the implementing agency.

Formats of the IFRs have been agreed with PCU and include (i) project sources and uses of funds, (ii) uses of funds by proposed project activity; (iii) DA statement, (iv) the disbursement summary, and (v) cash forecast, but might be updated with additional reports if required by other IFIs.

The PCU will consolidate the project financial information from all the PIUs using the online reporting system and produce full sets of quarterly IFRs to be submitted to the Bank within 45 days of the end of each calendar quarter, from the first disbursement and throughout the proposed project life.

The audit plans are established individually in each implementing agency and therefore the Bank is not able to solely rely on such audits for its fiduciary purposes. The audits of the proposed project financial statements will be conducted annually by independent auditors (with defined terms of reference) and according to auditing standards acceptable to the Bank.

For the ongoing Odra River Basin Flood Protection Project the financial audit has been performed by NIK and current auditing arrangements are satisfactory to the Bank. The annual audits of the proposed project financial statements will be provided to the Bank within six months after the end of each fiscal year, and also at the project closing.

The borrower will disclose the audit reports for the proposed project within one month of their receipt from the auditors and acceptance by the Bank, by posting the reports on its websites.

The MoF will open a separate Dedicated Account (DA) in the National Bank of Poland.

Withdrawal applications together with IFRs documenting the funds utilized from the DA and cash forecast requirement will be sent to the Bank every three months.

To facilitate the internal administration of financial flows, the share of the IBRD funding in the overall proposed project cost would be determined on an annual basis, through an Annual Work Plan (AWP) prepared by the Borrower, and agreed between the Borrower, the Bank and other partners involved in the proposed project implementation. AWP would provide a list of all annual activities supported by the proposed project and the co-financing arrangements for the relevant year.

However, for contracts below US\$20 million per contract, if the PIUs follow the NCB procedure allowing use of an open tender under the Polish Procurement Law, the Bank will review and agree on appropriate SBD.

The proposed project would further
Procurement of goods and non-consulting services.
advanced equipment and mathematical simulation models.

Selection of con

nsultants.

The consultant services under the proposed project will include,

Flood Protection of Middle and Lower

Odra

1.

Improvement of the Flood Monitoring and Warning System								
Telemetric Weather Monitoring Stations	155.0	50.0	205.0	37.3	12.0	49.3	24	5
Hydrometeorological Flood Forecasting	11.2	3.6	14.8	2.7	0.9	3.5	24	-
Modernization of POLRAD Network	27.9	9.0	36.9	6.7	2.2	8.9	24	1
Modernization of Hydrology Systems	22.6	7.3	29.9	5.4	1.8	7.2	24	1
Integration of Hydrological and Hydraulic Modelling	28.1	9.1	37.1	6.7	2.2	8.9	24	1
Nowcasting System and Flash Flood Early Warning	27.9	9.0	36.9	6.7	2.2	8.9	24	1
Subtotal	272.6	87.9	360.5	65.5	21.1	86.7	24	8
2.								

IB
Am

I. Investment Costs	
A. Civil Works	2
B. Land Acquisition and Resettlement Costs	
C. Design Costs	
D. Supervision of Construction	
E. Studies	
F. Goods	
G. Operating Costs	
Total PROJECT COSTS	2
Front-end fees	
Total Disbursement	2

The quarterly reports would cover, among others, the progress and expected completion date for civil works and goods contracts (both physical and financial progress with photographic evidence), progress and expected completion date for key consultancies, compliance towards environment and social safeguards, including implementation of key features of the environmental management plan (EMP) and resettlement action plans (RAPs), progress on institutional components, progress toward indicators given in the results framework, training and studies, and activities of the PCU consultants. The reports would also cover detailed financial and procurement information, including (i) a comparison of actual physical and financial outputs with forecasts, and updated six-month project forecasts; (ii) project financial statements, including sources and application of funds, expenditures by category statement, and special accounts reconciliation statement; and (iii) a procurement management report, showing status and contract commitments.

The frequency of these monitoring tables will be determined as needed.

Later, the Bank will field semiannual supervision missions. A mid-term review mission will be fielded in 2018/19, with the duration depending on the progress in implementation and the possible emergence of structural implementation challenges, the resolution of which would require an in-depth analysis and restructuring.

The project supervision missions would occasionally be integrated with missions of the other ongoing Odra River Basin Flood Protection Project under the same Steering Committee. The project procurement, FM, and safeguards missions could be undertaken as part of the periodic fiduciary support done concurrently across operations of all Bank sectors in Poland (the Procurement Specialist and FM specialist are based in the country unit, and the environmental specialist and social safeguards specialist in other offices in Europe).

As part of its project implementation support and supervision missions, the FM Supervision project effectiveness and then at appropriate intervals.

As required, a Bank-accredited FM specialist will participate in the implementation support and supervision process.

The Bank team will assist the PCU in drafting the terms of reference of the M&E consultant and the specifications for procuring any monitoring equipment.

All EMPs will be included in the respective bidding documents, both for construction and supervision.

The existing flood protection consists of passive dike systems of which the crest height was calculated to withstand high-water waves smaller than the one that was calculated to occur once in 100 years (1% chance of occurrence).

As the basins have evolved into economic growth the increasing economic damages now warrant the costs of increased protection levels.

Low temperatures in winter aggravate flood hazards and require control of ice-jams through an effective icebreaking operation to cut off the ice floes affecting the commercial Odra navigation.

The Project would play a primary role in reducing the frequency and severity of flooding in the project area. The project would provide three distinct areas with flood management infrastructure and related measures: (i) the Lower and Middle Odra; (ii) the Nysa Klodska valley, a large sub-basin of the Upper Odra; and (iii) the Upper Vistula. The population in the proximity of the proposed works and measures is estimated to be about 15.1 million, of which about 5.2 million are directly exposed to flood-related impacts, physically, economically and/or socially. The project would also further strengthen the national flood forecasting and operational capability (for existing and new infrastructure such as locks, weirs, barrages, reservoirs, etc).

A fundamental part of the economic appraisal is the estimation of the expected benefits to be attained by the project investments.

The reduced damages from floods were estimated based on the expected value of avoided Annual Average Damages (AAD) determined by the integration of a series of single event damages for a sequence of floods with progressively infrequent return periods. Maps of flooded areas were drawn separately for each flood risk area. Flood Risk Management Plans, co-funded by the EU under the Operational Program Technical Assistance 2007-2013, were useful in modeling the flooded areas. Flood hazard areas were determined by water table ordinates, resulting from a mathematical hydraulic modeling using GIS, based on a digital terrain model (DTM). Flood hazard areas are presented as surface objects, with an assigned value of the likelihood of flooding.

The assessment was based primarily on materials developed in the ISOK Project (IT System of the Country's Protection against Extreme Hazard^{di}), carried out by a consortium of governmental and academic institutions (i.e, the National Water Management Authority (KZGW); the Institute of Meteorology and Water Management - National Research Institute (IMGW-PIB); the Head Office of Land Surveying and Cartography; the National Institute of Telecommunications - National Research Institute; and the Government Center for Security.)

The estimates are based on the actual cost of construction of similar works of proper quality. In local currency the estimated costs amounts to PLN 4.74 billion, for the activities until project closure by 2023.

These costs are estimated at 0.5 percent of the value of the construction works which is the target value. This exceeds the percentage amounts made available so far from the Polish budget for maintaining the flood management structures. As part of the annual work program planning process, the Borrower will include O&M allocations to ensure these levels.

Depending on the asset registration, the annual recurrent budget for O&M is to be secured from the budgets of the RZGWs under the Ministry of Environment, or from the budget of the relevant ZMiUW, under the respective Regional Government (Voivodship). The ZMiUWs, in addition, receive earmarked budget for O&M from the State, i.e., from the Ministry of Agriculture, and from the Disaster Recovery Fund, Ministry of Administration and Digitization.

These budgets are prioritized to those assets that were financed by the Borrower EU funds; World Bank-funded assets would also receive this priority. The Early-warning system is funded partly under the IMGW budget and partly under the budget of the respective RZGWs.

Given the increasing frequency of flood hazards, the government, over the years, is also increasing the budget allocations to better operate and maintain the flood protection infrastructure. As the economy is expected to continue its growth, the additional fiscal burden to maintain and operate the new infrastructure from the time of completion of the works around 2020 or thereafter, can be, in principle, well accommodated in the growing budgets of the Implementing Agencies.

The Flood Risk Management Plans would include incentives to induce higher budgets for this purpose.

For component 4, the early warning system (EWS), the annual operating cost was assumed at 2 percent of the investment in equipment and construction works. Also, the additional salary costs were included for running the new EWS system, together with charges for up to 20 people to be employed in the service departments and hydrological forecasting bureaus.

For the economic analysis, value added tax (VAT) and price contingencies were excluded from the costs, and a standard conversion factor of 0.9 was applied.

Flood losses were calculated for each of the estimated areas falling under the flood risk maps.

The actual damages of the 1997, 1998 and 2010 floods, together with affected land use classes were used to derive unit rates of flood damage in values per square meter or per ha of each land use class to establish expected A Hydrodynamic simulation models were developed to provide water levels for floods with their respective return periods. Flood maps were then overlaid on the topography derived from digital terrain models, to estimate inundated areas and damages in simulated scenarios.

For residential and industrial areas loss values were obtained by comparing the value of assets with the degree of loss of assets dependent on the depth of the flood.

As the values of assets in the Regulation are presented in 2008 prices, the values were updated based on the corresponding index according to the Central Statistical Office (GUS).

³ Regulation of the Minister of the Environment, the Ministry of Infrastructure and the Minister of Internal Affairs and Administration on the development of flood hazard maps and flood risk maps.

Agricultural production losses were estimated considering the existing cropping pattern of the respective area. The average revenue per crop (in PLN/ha) and the crop area for each province were based on data of the GUS in the period 2011-2014, including average prices of products, crop yields, crop areas, surface of arable land, etc. The weighted average crop values for each area were estimated taking 2014 prices. Assessment of the net financial effect of floods on crop yields and other agricultural activities during flood seasons allowed valuing the flood effects on these activities. Variable crop production expenditures were used in the assessment of losses valued at farm gate and adjusted to world prices to reflect the economic loss due to flooding.

This estimation was taken as the starting point for valuing the non-material damages for this assessment, adjusting the methodology to the Polish setting³.

The 'loss of life models', which include the characteristics of a given flood and the population at the risk of flooding, are also being worked out (Jonkman, 2002).

³ Estimates developed for the Feasibility Study for the Modernization of the Wrocław Floodway System in 2012 were followed. Willingness of the British to bear costs of avoiding flood effects was corrected by using the GDP per capita in Great Britain and Poland. The estimated Polish value for avoiding the effects of flooding (amounted to about PLN 388 per household in 2004 values) and was adjusted for inflation.

Other quantified benefits from the proposed project are those related with the Keynesian multiplier effect.

The induced benefits were estimated using this multiplier and the average net profit for the Polish business investments which was estimated at 4.6 percent in the period from January to June 2011.

The proposed investment will also have an enabling environment for increasing the tourism income flowing into some of the proposed protected areas, as the risk of floods is reduced and the safety for tourists is increased.

These benefits were quantified taking into account the current number of tourists per area and the cost of stay per day for the average traveler (approximately PLN 90^{bn}).

The proposed repair and improvement of river training structures on the border section of the Odra downstream to the estuary will enhance navigation conditions by making more uniform the water flow and transport conditions, as a consequence of measures to reduce the ice-jamming. The restoration of river training structures and raising of five bridges will enhance navigation conditions, increasing its safety and competitiveness. The Odra has a direct connection with the waterway system in the EU which could be conducive to reducing transport costs in trade exchange between Poland and EU countries.

For quantifying this benefit it was assumed that cargo transported on the Odra River (containers, construction materials, grains, etc.)

Benefits of PLN 32.17 million per year were estimated by multiplying the expected increase of transported cargo by the reduction in transport costs per ton/km through the river^{et}.

The average cost of road transport was estimated at an average of PLN 128.75 per ton-km, while the waterway transport cost is on average PLN 22.26.

Exploitation of aggregate mining. The Project is expected to generate also benefits from the exploitation of gravel and other materials for construction in the proposed reservoir areas. It is estimated that around 4 Mm³ of gravel would be potentially available annually for extraction from the reservoirs valued at about PLN 100 million of gravel extraction (in period 2016-2027).

Hence, benefits of EWS are reduced material and non-material losses and induced economic benefits.

This predicts damage reduction in terms of percentage of maximum potential inundation damage as a function of the mitigation time.

The assessment was carried out by comparing the incremental capital and operating costs required with the expected incremental benefits expected to result from the implementation of the project. The evaluation considered a 50-year period and two different alternatives for the estimation of the results: using a discount rate of (i) 5 percent (following the Guide to Cost-Benefit Analysis of Investment Project, European Commission, 2008, as well as OECD guidance); and (ii) 10 percent (as the minimum usually applied by the World Bank for this type of analysis). A standard conversion factor of 90 percent was used for conversion from financial to economic values.

The economic feasibility of the project and of each of its main components was measured through estimation of the economic rate of return (ERR) and net present values (NPVs) based on the above described assumptions. As can be seen in Table 5.1, the estimated overall project ERR is 13.5 percent, which is well in excess of the long term opportunity cost of capital (OCC). The analysis also showed that all investments proposed for the project areas would have an impact that justifies the costs involved in the proposed studies. The experience with the ongoing Odra River Basin Flood Protection Project has shown that the government is committed to provide the budget for investment as well as for proper O&M. The field inspections in the territories of the prospective PIUs of the project demonstrate that, in general, modest but reasonable resources are provided for the maintenance and functioning of the infrastructure, recognizing that current O&M budget levels cannot compensate for the funding shortages in the second half of the 20th century.

A sensitivity analysis was carried out for the entire project for checking the strength of the economic performance indicators to changes in critical assumptions of the analysis.

The following assumptions were selected for testing variations and the sensitivity of the expected results: (i) investment costs; (ii) O&M costs; (iii) benefits from material and non-material avoided-flood damages; and (iv) the induced economic benefits.

The analysis shows that the estimated ERR is robust and not very sensitive to changes in costs and/or benefits: even a 30 percent increase in investment costs, or up to 40 percent reduction of expected benefits from avoided material flood damage would maintain the ERR above 10 percent.

Fiscal budgets for operation and maintenance (O&M) of flood protection structures are in Poland usually adequate but on the lower side of the common ranges Table 5.2 below shows, by Implementing Agency, the ratio of confirmed available budgets for O&M in calendar year 2015 to the respective value of assets.

Hence, taking into consideration the current practice concerning public funds allocations for this purpose in Poland, for the economic and financial analysis it has been assumed that the budgets for O&M would be about 0.5 percent of the value of assets.

The allocation of the public funds for O&M of flood protection structures depends on the policy guidance and levels proposed at the national level, which is subject to parliamentary analysis and negotiations every year. Several policy review documents and technical papers including economic analysis of sector projects co-funded by the International Financial Institutions recommend an increase of the current levels of funds available for O&M of the flood protection structures. The issue is attracting growing interest of national policy makers as well as awareness about the need for higher O&M allocations for existing and new facilities given the increased frequency and intensity of flood events.

The assumed 0.5 percent annual maintenance cost for the structures to be constructed under the proposed project amounts to only 0.007 percent of the total State Budget expenditure. Compared with the benefits expected to be generated by the protection investments in terms of avoided damages and other benefits, its incidence is very low being equivalent to about 4.7 percent of the benefits. It is expected that O&M allocations are due to slowly increase to further mitigate against risks from unexpected events. The fiscal impact of the investment and its recurrent costs is low.



